

Guided de-escalation of antiplatelet treatment in patients at high atherothrombotic risk presenting with acute coronary syndrome: a *post hoc* analysis of the TROPICAL-ACS trial

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Aims

Platelet function testing (PFT)-guided de-escalation of dual antiplatelet therapy (DAPT) after acute coronary syndrome (ACS) has been shown to reduce bleeding risk without increasing ischaemic events. Patients with high atherothrombotic risk (ATR) remain a challenging subgroup for such strategies. This study evaluated the safety and efficacy of early PFT-guided DAPT de-escalation according to ATR status.

Methods and results

In the TROPICAL-ACS trial, 2610 ACS patients were randomized to standard 12-month prasugrel therapy or PFT-guided DAPT de-escalation. For this *post hoc* analysis, patients were stratified by ATR. High ATR was defined as age ≥ 65 years, polyvascular disease, or ≥ 2 risk factors (diabetes, smoking, or renal dysfunction). Hazard ratios (HRs) for clinical endpoints were derived using multistate Cox regression. High-ATR patients ($n = 990$) had a higher incidence of the primary net clinical benefit endpoint—composite of cardiovascular death, myocardial infarction, stroke, or Bleeding Academic Research Consortium 2–5 bleeding—compared with low-ATR patients ($n = 1620$) (11.0% vs. 6.7%; HR 1.67, 95% confidence interval 1.28–2.18; $P < 0.001$). Platelet function testing-guided de-escalation showed no significant interaction by ATR status for the primary endpoint [high ATR: 10.5% vs. 11.5%, HR 0.90 (0.61–1.32), $P = 0.586$; low ATR: 5.6% vs. 7.7%, HR 0.71 (0.48–1.04), $P = 0.082$; $P_{\text{interaction}} = 0.394$] or ischaemic events [high ATR: 3.7% vs. 4.4%, HR 0.83 (0.44–1.56); low ATR: 1.8% vs. 2.6%, HR 0.68 (0.35–1.34); $P_{\text{interaction}} = 0.666$].

Conclusion

Early PFT-guided de-escalation from prasugrel to clopidogrel was safe across atherothrombotic risk categories, supporting individualized DAPT optimization in ACS patients with both low and high ischaemic risk.

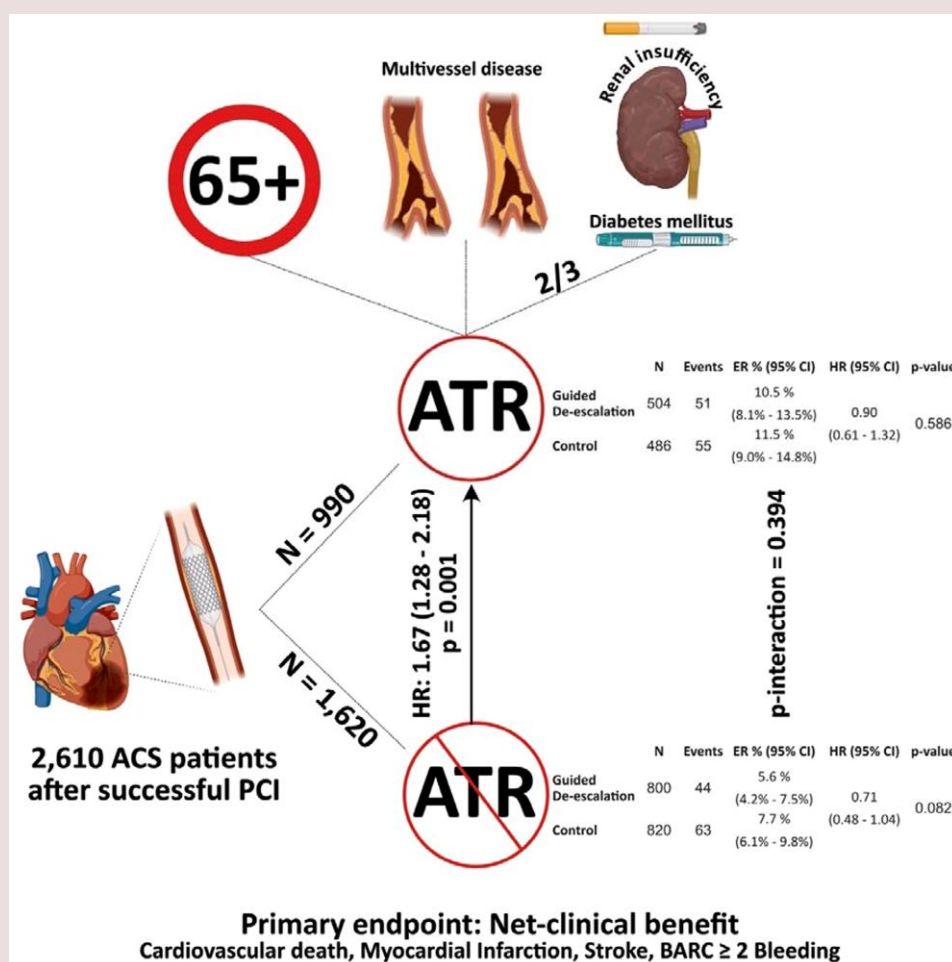
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Graphical abstract



Graphical abstract summarizing the main findings of this study. ACS, acute coronary syndrome; ATR, atherothrombotic risk; BARC, Bleeding Academic Research Consortium; ER, event rate; HR, hazard ratio; PCI, percutaneous coronary intervention. Created with BioRender.com.

Keywords

Acute coronary syndrome • Diabetes • Elderly • Renal insufficiency • Multivessel vascular disease • Adjunctive pharmacotherapy

Introduction

Patients with an acute coronary syndrome (ACS) and undergoing percutaneous coronary intervention (PCI) are at high risk for subsequent ischaemic as well as bleeding events.¹⁻⁴ Dual antiplatelet therapy (DAPT) is associated with decreased ischaemic events at the expense of increased bleeding risk, especially when potent P2Y₁₂ inhibitors are used.^{3,4} The risk of ischaemic and bleeding events is subject to an individual risk pattern and temporal changes after the initial ACS, with a higher risk for ischaemic events in the acute phase and an increasing risk for bleeding in the chronic phase.⁵⁻⁷ This observation led to the development of strategies using potent antiplatelet drugs for the acute phase followed by de-escalation to clopidogrel in the maintenance phase after an ACS. In the TROPICAL-ACS trial, a platelet function testing (PFT) guided early de-escalation of prasugrel to clopidogrel proved to be non-inferior to the current standard of care, i.e. prasugrel for the first 12 months of PCI for ACS.⁸ A tailored escalation or de-escalation is desired for the individual patient depending on the risk for thrombosis or bleeding.⁹ Stratifying

patients according to their individual atherothrombotic risk (ATR) is now encouraged by latest guidelines and reflects modern clinical practice.^{10,11} As a recent example, the large-scaled randomized COMPASS (Cardiovascular Outcomes for People Using Anticoagulation Strategies) trial focused on patients with chronic coronary syndromes (CCSs) and used a distinct definition to select patients with elevated ATR.¹² Patients with high ATR constitute an especially challenging population regarding decisions on DAPT de-escalation, and therefore dedicated analyses are mandatory. Thus, in this exploratory *post hoc* analysis of the TROPICAL-ACS trial, we aimed to investigate the safety and efficacy of PFT-guided P2Y₁₂ de-escalation from prasugrel to clopidogrel in patients with low vs. high ATR.

Methods

Anonymized data, underlying this article, will be shared on reasonable request to the corresponding author. The design, in- and exclusion criteria, and modality of PFT as well as results of the TROPICAL-ACS main trial

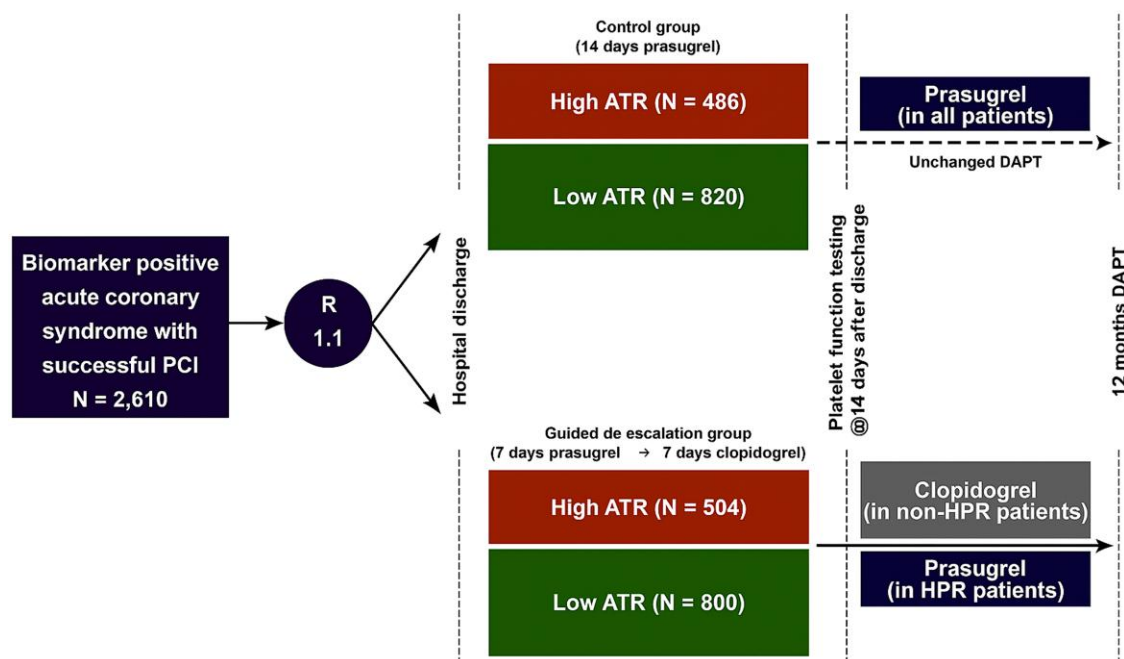


Figure 1 Study design and groups. Patients were randomly assigned (1:1) after percutaneous coronary intervention and directly before planned discharge from hospital to a standard treatment with prasugrel for 12 months or to platelet function testing-guided dual antiplatelet therapy de-escalation. For this specific *post hoc* and exploratory analysis, patients within the trial were stratified according to their atherothrombotic risk profile. High atherothrombotic risk was defined as age ≥ 65 years, polyvascular disease, or presence of at least two of the following risk factors: diabetes mellitus, smoking, or renal dysfunction. ATR, atherothrombotic risk; HPR, high platelet reactivity; PCI, percutaneous coronary intervention.

have been described previously in detail.^{8,13} The trial was an investigator-initiated, randomized, parallel-group, open-label assessor-blinded multicentre trial including all-comer ACS patients undergoing PCI and was conducted in 33 European countries (ClinicalTrials.gov identifier: NCT01959451). In brief, patients with successful PCI for biomarker-positive ACS were 1:1 randomized prior to discharge between two different strategies of DAPT (Figure 1). Antiplatelet regimens consisted of either a PFT-guided de-escalation strategy or the standard treatment strategy with prasugrel—both in combination with aspirin over a period of 12 months after PCI. Patients in the PFT-guided DAPT de-escalation group received a 1-week post-discharge treatment with prasugrel 10 or 5 mg per day (according to patient characteristics) followed by 1-week treatment with clopidogrel (75 mg/d) and a PFT at the end of the second week (guided de-escalation group). Based on PFT results in the guided de-escalation group, patients were either switched back to prasugrel, in case insufficient platelet inhibition was detected [= high platelet reactivity (HPR)], or continued with clopidogrel over a total period of 12 months post-PCI.⁸ Patients in the control group were planned to receive a 12 months treatment with prasugrel 10 or 5 mg according to current guidelines.¹³ In the TROPICAL-ACS trial, we used the Multiplate analyser (Roche Diagnostics, Rotkreuz, Switzerland) for PFT. High platelet reactivity was defined based on the results of previous studies and the consensus documents of the Working Group on HPR as an adenosine diphosphate test aggregation value of 46 units or higher on the Multiplate analyser.^{14,15} Patients with HPR immediately switched back to prasugrel, while patients without HPR continued clopidogrel. The present analysis included the intention-to-treat population. The study was approved by the local ethics committees of each participating site, as well as by the competent national agencies as required. It was conducted in agreement with the Declaration of Helsinki.

Stratification according to the atherothrombotic risk profile

For this specific *post hoc* and exploratory analysis, patients within the trial were stratified according to their ATR profile. A two-group categorical

classification separated the total TROPICAL-ACS collective into patients with low vs. high ATR. The recently published ATR classification strategy from the original COMPASS trial¹² was adapted to the TROPICAL-ACS patient collective. In line with COMPASS definitions,¹² high-ATR patients were required to meet one of the following criteria:

- (1) Age ≥ 65 years
- (2) Age < 65 years and atherosclerosis involving at least two vascular beds
- (3) Age < 65 years and presence of at least two of the following risk factors:
 - Diabetes mellitus (DM)
 - Current smoking
 - Renal dysfunction (estimated glomerular filtration rate < 60 mL/min)

Patients not fulfilling any of those criteria were defined as low ATR. Figure 1 illustrates the study patients from TROPICAL-ACS according to study groups and stratified by ATR.

Study endpoints

Clinical endpoint analyses for this specific study were conducted in line with endpoint assessments for the primary trial results.^{8,13} In brief, the primary endpoint was a combined ischaemic and bleeding endpoint (net clinical benefit) consisting of a composite of cardiovascular death (CVD), myocardial infarction (MI) defined according to the 3rd universal definition of MI,¹⁶ stroke, or bleeding Type 2 or higher according to the Bleeding Academic Research Consortium (BARC) criteria¹⁷ at 12 months after randomization. Further endpoint assessments included BARC 2–5 bleeding events (key secondary endpoint); the composite of CVD, MI, or stroke; and individual components of the above-mentioned endpoints.

Table 1 Baseline characteristics of the study cohort in patients with high and low atherothrombotic risk

	Control group	High ATR (n = 990) Guided de-escalation	P-value	Control group	Low ATR (n = 1620) Guided de-escalation	P-value
n	486	504		820	800	
Age, years	66.9 (8.4)	67.6 (7.8)	0.241	53.4 (7.5)	53.5 (7.2)	0.983
Male sex	360 (74.1)	378 (75.0)	0.794	663 (80.9)	651 (81.4)	0.838
Body mass index, kg/m ²	28.3 (4.7)	28.1 (4.4)	0.560	28.7 (5.3)	28.2 (4.6)	0.441
Caucasian race	480 (98.8)	502 (99.6)	0.243	815 (99.4)	793 (99.1)	0.561
Previous percutaneous coronary intervention	96 (19.8)	75 (14.9)	0.052	90 (11.0)	98 (12.2)	0.470
Previous coronary artery bypass surgery	29 (6.0)	28 (5.6)	0.888	17 (2.1)	11 (1.4)	0.375
Previous myocardial infarction	78 (16.0)	61 (12.1)	0.090	75 (9.1)	79 (9.9)	0.678
Previous peripheral artery occlusive disease	39 (8.0)	46 (9.1)	0.613	0 (0)	0 (0)	NA
History of coronary artery disease	108 (22.2)	89 (17.7)	0.086	96 (11.7)	86 (10.8)	0.595
History of renal insufficiency	29 (6.0)	27 (5.4)	0.781	5 (0.6)	6 (0.8)	0.967
Current smoker	166 (34.2)	168 (33.3)	0.836	425 (51.8)	423 (52.9)	0.710
Arterial hypertension	361 (74.3)	370 (73.4)	0.812	445 (54.3)	423 (52.9)	0.608
Hyperlipidaemia	225 (46.3)	254 (50.4)	0.220	304 (37.1)	292 (36.5)	0.851
Family history of coronary artery disease	156 (32.1)	139 (27.6)	0.138	310 (37.8)	280 (35.0)	0.262
Haemoglobin (g/dL)	13.9 (1.6)	14.1 (1.5)	0.022	14.5 (1.3)	14.4 (1.4)	0.784
Creatinine (mg/dL)	1.0 (0.3)	1.0 (0.4)	0.963	0.9 (0.2)	0.9 (0.2)	0.778
Medication at admission						
Aspirin	186 (38.3)	165 (32.7)	0.080	157 (19.1)	138 (17.2)	0.355
ADP receptor antagonist	40 (8.2)	32 (6.3)	0.309	36 (4.4)	39 (4.9)	0.729
Beta-blocker	194 (39.9)	190 (37.7)	0.515	174 (21.2)	188 (23.5)	0.297
ACE inhibitor	171 (35.2)	161 (31.9)	0.311	170 (20.7)	190 (24.5)	0.079
AT1 receptor antagonist	80 (16.5)	96 (19.0)	0.327	81 (9.9)	77 (9.6)	0.930
Calcium antagonist	87 (17.9)	106 (21.0)	0.245	79 (9.6)	72 (9.0)	0.724
PPI	88 (18.1)	85 (16.9)	0.667	87 (10.6)	89 (11.1)	0.800
Statin	160 (32.9)	146 (29.0)	0.202	138 (16.8)	140 (17.5)	0.770

ADP, adenosine diphosphate; ACE, angiotensin-converting enzyme; AT1, angiotensin 1; PPI, proton pump inhibitor; ATR, atherothrombotic risk.

Statistical analysis

Continuous variables are presented as means with standard deviations and were compared using the Wilcoxon rank-sum test. Categorical variables are expressed as percentages and were analysed using the χ^2 tests. All ATR-related analyses were performed on an intention-to-treat basis. Survival curves were estimated using competing-risk Kaplan–Meier analysis. The hazard ratios (HRs) and 95% confidence interval (CI) for all outcome measures were evaluated using semi-parametric multistate Cox-regression analysis.¹⁸ Non-cardiovascular mortality was defined as competing risk for the primary, secondary, ischaemic, and cardiovascular mortality (CVM) endpoints. All-cause mortality was defined as competing risk for bleeding endpoints, MI, stroke, stent thrombosis, and urgent revascularization. The multi-state model included the following states: (i) healthy state, (ii) primary or secondary endpoint, and (iii) competing risk (non-cardiovascular mortality or all-cause mortality), as well as the following state transitions (healthy state and then primary or secondary endpoint, (ii) healthy state and then competing risk (non-cardiovascular mortality or all-cause mortality), and (iii) primary or secondary endpoint and then competing risk (non-cardiovascular mortality or all-cause mortality). Differences in treatment effects, as well as subgroup analyses, were implemented using interaction testing. Non-inferiority testing for

the primary endpoint was tested using a one-sided 30% margin, as it was pre-specified in the protocol of the trial. Since there was a random allocation to treatment (guided de-escalation vs. control group), all analyses stratified by treatment were not adjusted for any baseline characteristics. For all analyses, differences were considered statistically significant when the two-sided *P*-value was <0.05. All statistical analyses were performed using CRAN R, versions 3.5.1 and 4.2.2.

Results

Study population

Between December 2013 and May 2016, 2610 patients were enrolled. Of these, 1304 were randomized to the PFT-guided DAPT de-escalation group and 1306 to the control group. Overall, 990 patients (37.9%) were stratified as high ATR (guided de-escalation 504 vs. control 486 patients) and 1620 (62.1%) as low ATR (guided de-escalation 800 vs. control 820 patients). The composition of the individual components of the high ATR is presented in [Supplementary material online, Table S1](#). Baseline clinical

Table 2 Angiographic and procedural characteristics of the study cohort stratified by atherothrombotic risk

	High ATR n = 990			Low ATR n = 1620		
	Control group	Guided de-escalation	P-value	Control group	Guided de-escalation	P-value
n	486	504		820	800	
PCI for			0.161			0.608
STEMI	230 (47.3)	262 (52.0)		492 (60)	469 (58.6)	
NSTEMI	256 (52.7)	242 (48.0)		328 (40)	331 (41.4)	
Access site			1.00			0.144
Brachial	0 (0)	0 (0)		3 (0.4)	0 (0)	
Femoral	201 (41.4)	209 (41.5)		340 (41.5)	314 (39.2)	
Radial	285 (58.6)	295 (58.5)		477 (58.2)	486 (60.8)	
No. of diseased coronary vessels			0.599			0.947
1	213 (43.8)	207 (41.0)		469 (57.2)	452 (56.5)	
2	133 (27.4)	151 (30.0)		212 (25.9)	208 (26.0)	
3	140 (28.8)	146 (29.0)		139 (16.9)		
Anticoagulant agent used for PCI			0.955			0.989
Bivalirudin	21 (4.3)	22 (4.3)		34 (4.1)	32 (4.0)	
Low molecular weight heparin	24 (4.9)	27 (5.4)		46 (5.6)	45 (5.6)	
Unfractionated heparin	441 (90.8)	455 (90.3)		740 (90.3)	723 (90.4)	
Use of GP IIb/IIIa antagonist	83 (17.1)	86 (18.1)	1.000	164 (20.0)	158 (19.8)	0.949
TIMI flow grade before PCI			0.367			0.879
0	170 (35.0)	192 (38.1)		342 (41.7)	319 (39.9)	
1	64 (13.2)	66 (13.1)		107 (13.0)	107 (13.4)	
2	117 (24.1)	130 (25.8)		185 (22.6)	191 (23.9)	
3	135 (27.7)	116 (23.0)		186 (22.7)	183 (22.8)	
Coronary vessel treated			0.008			0.330
LM	5 (1.0)	19 (3.8)		7 (0.9)	10 (1.2)	
LAD	202 (41.6)	214 (42.5)		345 (43.2)	348 (43.5)	
LCX	93 (19.1)	98 (19.4)		160 (19.5)	168 (21.1)	
RCA	164 (33.7)	164 (32.5)		286 (34.8)	169 (33.6)	
Coronary bypass graft	22 (4.6)	9 (1.8)		13 (1.6)	5 (0.6)	
AHA/ACC classification of lesions			0.907			0.873
A	55 (11.3)	53 (10.5)		100 (12.2)	108 (13.5)	
B1	148 (30.5)	163 (32.3)		277 (33.8)	195 (24.4)	
B2	133 (27.4)	132 (26.2)		207 (25.2)	195 (24.4)	
C	150 (30.8)	156 (31.0)		236 (28.8)	226 (28.2)	
Ostial lesion	39 (8.0)	46 (9.1)	0.613	59 (7.2)	51 (6.4)	0.577
Bifurcation lesion	70 (14.4)	86 (17.1)	0.289	125 (15.2)	118 (14.8)	0.835
Stent type						
DES	379 (78.0)	395 (78.4)	0.943	639 (77.9)	620 (77.5)	0.883
BMS	90 (18.5)	101 (20.0)	0.599	146 (17.8)	146 (18.2)	0.866
BVS	31 (6.4)	20 (4.0)	0.116	52 (6.3)	48 (6.0)	0.855
PTCA only	6 (1.2)	3 (0.6)	0.469	7 (0.9)	6 (0.8)	1.000

PCI, percutaneous coronary intervention; STEMI, ST-elevation myocardial infarction; NSTEMI, non-ST-elevation myocardial infarction; LM, left main; LAD, left anterior descending artery; LCX, left circumflex; RCA, right coronary artery; DES, drug-eluting stent; BMS, bare metal stent; BVS, bioabsorbable scaffold; PTCA, percutaneous transluminal coronary angioplasty; ATR, atherothrombotic risk.

characteristics are presented in [Table 1](#). Patients from high ATR were older and less frequently males and had higher rates of previous PCI, coronary artery bypass graft surgery (CABG), MI, and peripheral artery

disease compared with patients from low ATR ([Table 1](#)). Furthermore, high-ATR patients more frequently had a history of chronic kidney disease (CKD), hypertension, and hyperlipidaemia compared with

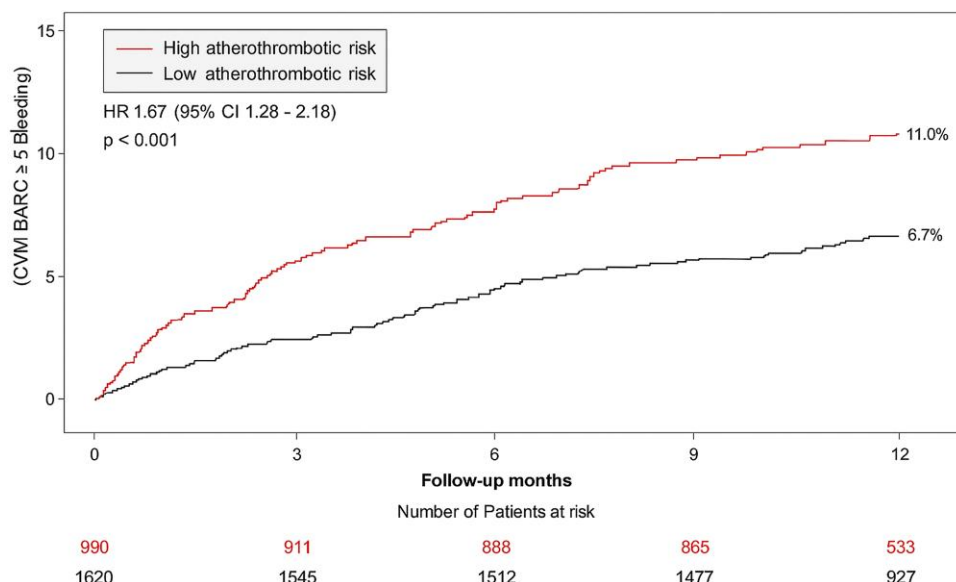


Figure 2 Kaplan–Meier curves for the primary endpoint stratified by atherothrombotic risk at 12 months of follow-up. Patients were stratified according to their atherothrombotic risk profile. BARC, Bleeding Academic Research Consortium; HR, hazard ratio.

low-ATR patients. The angiographic characteristics of groups are shown in [Table 2](#). Patients from high ATR had comparable angiographic and procedural characteristics in comparison to low ATR, with exception of significantly higher left main coronary artery treated in the high-ATR and guided de-escalation group ([Table 2](#)). There was no difference in the rate of patients who required a re-start of prasugrel based on PFT in both ATR groups (see [Supplementary material online, Table S2](#)).

Clinical endpoint distribution according to atherothrombotic risk stratum

The impact of the individual components of the high-ATR score and their impact on the primary endpoint and the combined ischaemic endpoint are presented in [Supplementary material online, Tables S3 and S4](#). Age > 65 years and DM were significant predictors of the primary endpoint in multivariate analysis. Diabetes mellitus and CKD were independent predictors of the combined ischaemic endpoint in multivariate analysis. The primary endpoint occurred more frequently in patients at high ATR compared with those at low ATR [11.0% vs. 6.7%, HR 1.67 (95% CI 1.28–2.18); $P = 0.001$] ([Figure 2](#)). High- and low-ATR patients showed similar rates of the primary endpoint when comparing the PFT-guided de-escalation subgroup with the control subgroup [10.5% vs. 11.5%; HR 0.90 (95% CI 0.61–1.32); $P = 0.586$ in the high-ATR stratum and 5.6% vs. 7.7%; HR 0.71 (95% CI 0.48–1.04); $P = 0.082$ in the low-ATR stratum]. No significant interaction between treatment strategy and ATR was found ($P_{\text{interaction}} = 0.394$) ([Figure 3](#)). With regard to the combined ischaemic endpoint, the guided de-escalation group compared with the control group showed similar event rates (3.7% vs. 4.4%; HR 0.83 (95% CI 0.44–1.56); $P = 0.567$ in the high-ATR stratum and 1.8% vs. 2.6%; HR 0.68 (95% CI 0.35–1.34); $P = 0.262$ in the low-ATR stratum; $P_{\text{interaction}} = 0.666$; [Figure 4](#)). Results were consistent across all pre-defined endpoints from the TROPICAL-ACS trial ([Figure 5](#)). High-ATR criteria were predictive of the primary endpoint and combined ischaemic as well as bleeding endpoints (BARC 2–5) ([Supplementary material online, Table S5](#)).

Discussion

To the best of our knowledge, this exploratory *post hoc* analysis from the randomized TROPICAL-ACS trial is the first investigation on the efficacy and safety of DAPT de-escalation from prasugrel to clopidogrel in ACS patients stratified by their individual ATR. Key findings of this analysis are as follows. First, in an all-comer cohort of invasively managed ACS patients, we found a substantial proportion of patients (~40%) that had a high ATR according to contemporary criteria.¹² Second, those patients with high ATR had worse clinical outcomes, reflected by an increased risk of combined ischaemic events and thereby rendering our cohort as a true high (atherothrombotic)-risk cohort. Third, PFT-guided de-escalation of DAPT from prasugrel to clopidogrel is safe regardless of patients' ATR. Fourth, the utilized high-ATR criteria were predictive of both ischaemic and bleeding endpoints.

Dual antiplatelet therapy including a potent P2Y₁₂ inhibitor may not be feasible or safe in various clinical scenarios (intended major surgery, socioeconomic factors, and side effects) and for certain patient subsets (elderly and patients with high bleeding risk). Several strategies to minimize the risk of bleeding without increasing ischaemic events have been investigated.^{19–21} TROPICAL-ACS has shown that guided de-escalation may be considered for these patients.⁸ Since the publication of these data, early guided de-escalation of DAPT has been recommended in current guidelines and consensus documents on ACS and DAPT management.^{11,22} Recently, the TROPICAL-ACS trial data has been scrutinized for various patient subsets, and those analyses indicated that certain sub-collectives seem to profit more than others from a guided de-escalation strategy.^{23–27} In this regard, assessing the safety and efficacy of TROPICAL-ACS strategy by stratification of patients into clusters of low and high ATR seems reasonable. Patients in the group with high ATR were older, had more frequently chronic comorbidities with respective medication, and were more frequently treated for non-ST-elevation myocardial infarction (NSTEMI)-ACS and multivessel disease. Along with the findings of other dedicated studies on patients at moderate to high ATR,^{12,28–31} an increase in ischaemic as well as bleeding events was observed in the current analysis. Despite differences to earlier studies in the selection of variables to define ATR,

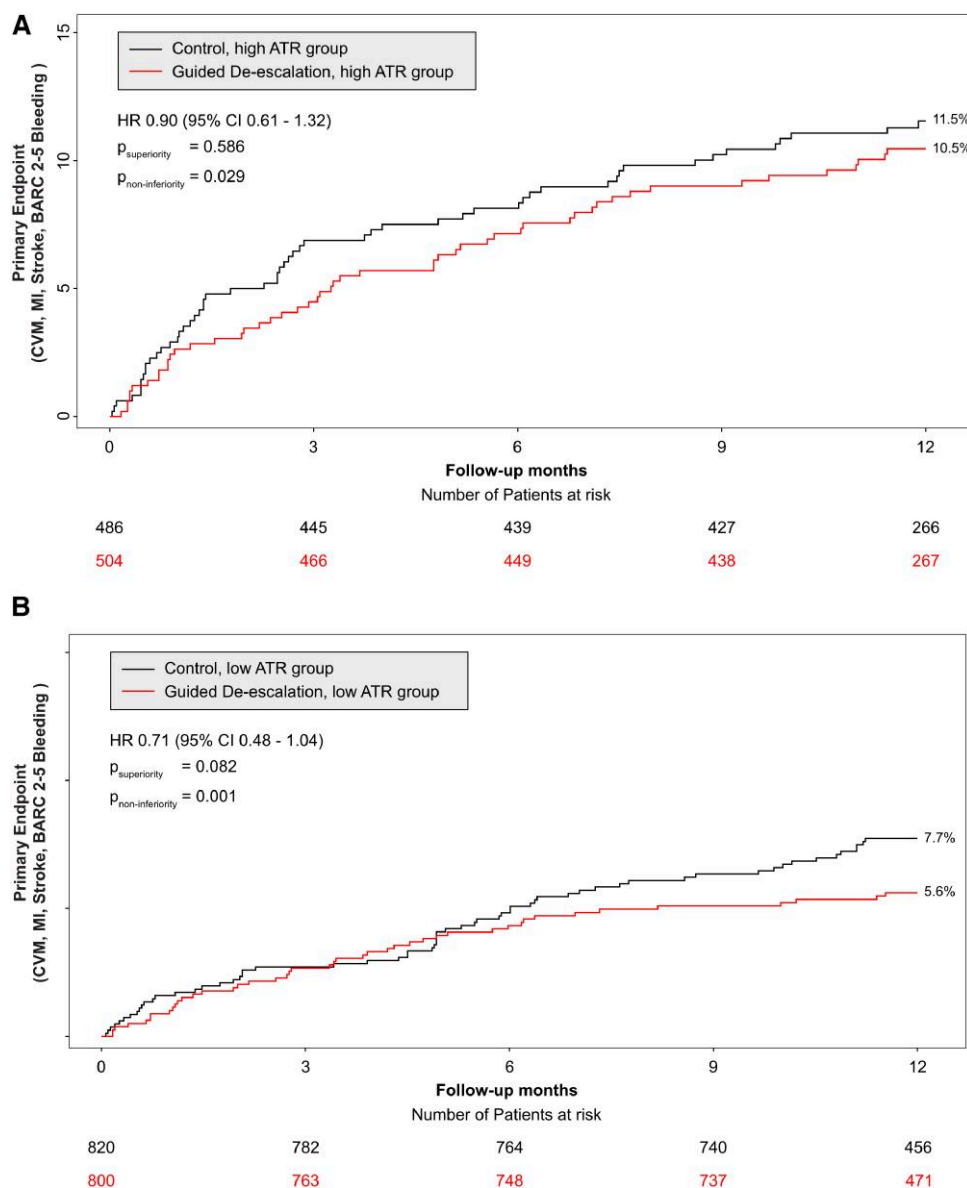


Figure 3 Kaplan–Meier curves for the primary endpoint stratified by treatment strategy at 12 months of follow-up. (A) High atherothrombotic risk was defined as age ≥ 65 years, polyvascular disease, or presence of at least two of the following risk factors: diabetes mellitus, smoking, or renal dysfunction. (B) Low atherothrombotic risk. ATR, atherothrombotic risk; BARC, Bleeding Academic Research Consortium; HR, hazard ratio.

the criteria applied in the present analysis showed a strong capability to discriminate between collectives at low and significantly increased the risk for cardiovascular events. Though the COMPASS inclusion criteria were not tested and implemented specifically in a post-ACS setting and not designed to filter patients at the highest ATR, it may be used to separate patients at low from those at moderate and high risk for future cardiovascular events. Other risk scores with similar components were implemented and investigated in a wide variety of patients (the PARIS score³² and TIMI Risk Score for Secondary Prevention, TRS-2P³³). The TRS-2P score, composed of nine components (age ≥ 75 , congestive heart failure, hypertension, DM, prior stroke, prior CABG surgery, peripheral arterial disease, current smoking, and renal dysfunction), showed a strong association with subsequent events after a MI in a sub-study of the TRA 2°P–TIMI 50 trial (Thrombin Receptor

Antagonist in Secondary Prevention of Atherothrombotic Ischemic Events–TIMI 50).³³ Additionally, the COMPASS criteria appear to predict both ischaemic and bleeding events as depicted in [Supplementary material online, Table S5](#). This finding might appear surprising at first glance. Several features of the High Bleeding Risk Academic Research Consortium (HBR-ARC) such as age and CKD are well-known factors to increase the risk of bleeding in patients after PCI.³⁴ Similarly, DM can be considered a feature of increased bleeding risk.³⁵ Nevertheless, the impact of ATR criteria of the COMPASS on bleeding risk requires further investigation from prospective studies.

Key finding from this analysis is that the TROPICAL-ACS strategy of PFT-guided early P2Y₁₂ de-escalation compared with control group is not associated with an increase in ischaemic nor bleeding events, regardless of ATR. It was reassuring to see that none of the predefined

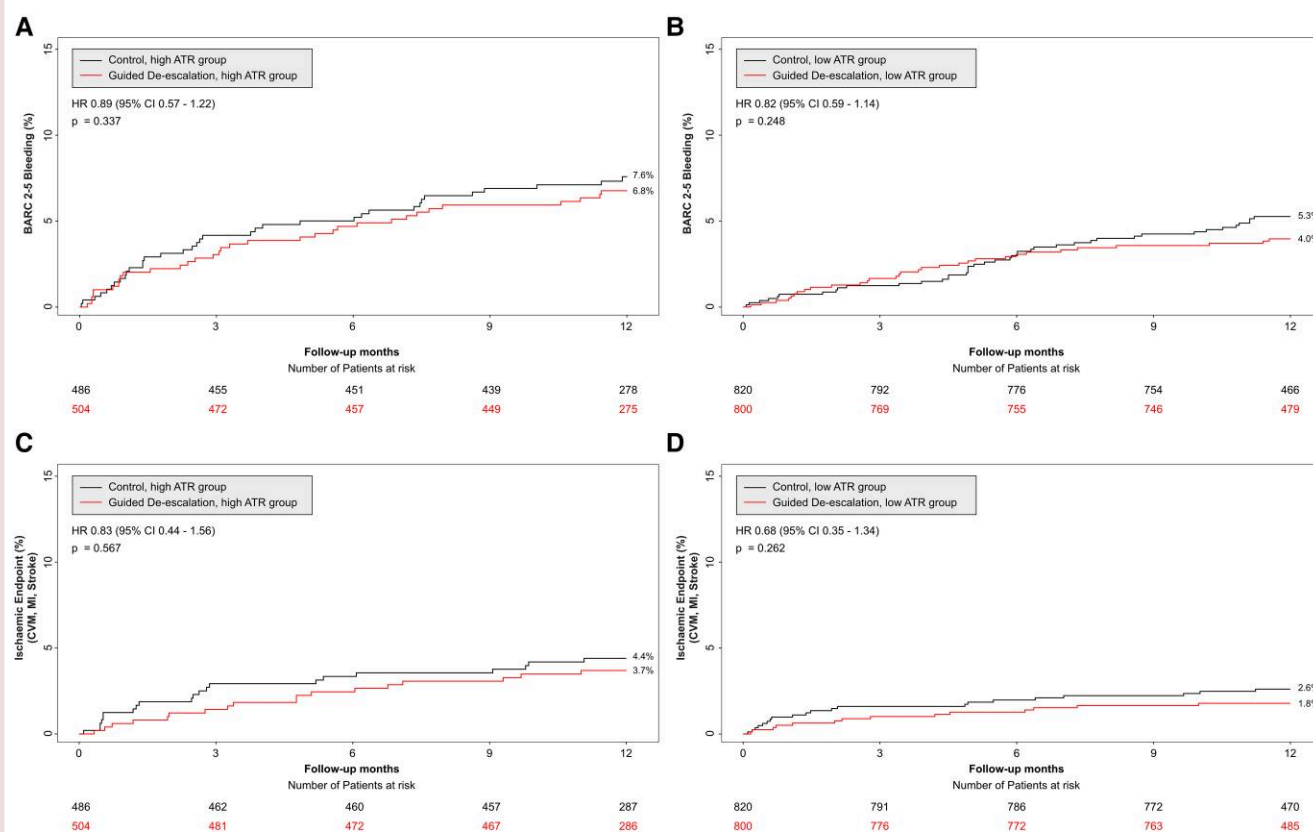


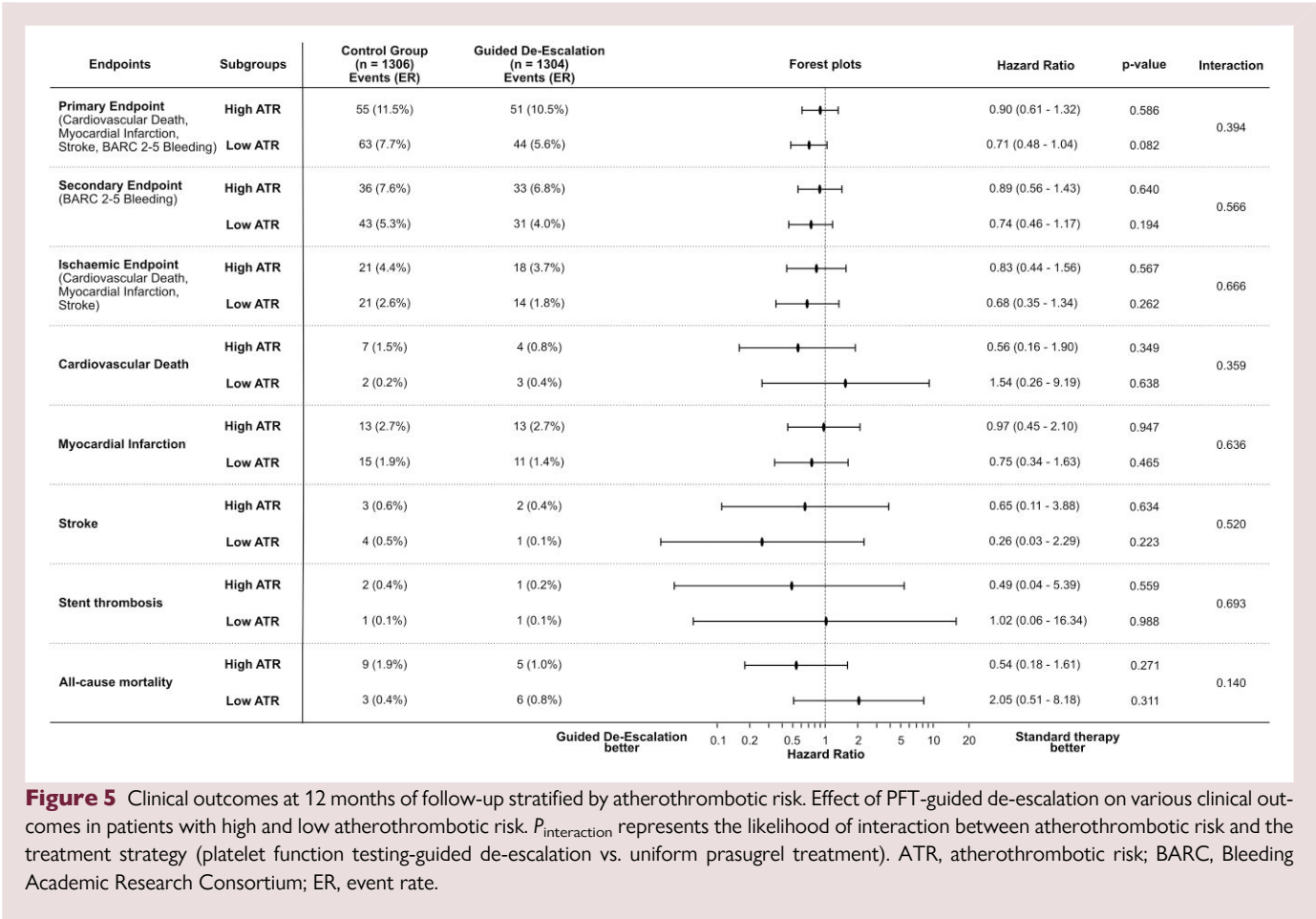
Figure 4 Kaplan–Meier curves for the secondary and ischaemic endpoints in patients stratified by treatment strategy at 12 months of follow-up. (A) Secondary endpoint in patients with high atherothrombotic risk, defined as age ≥ 65 years, polyvascular disease, or presence of at least two of the following risk factors: diabetes mellitus, smoking, or renal dysfunction. (B) Secondary endpoint in patients with low atherothrombotic risk. (C) Ischaemic endpoint in patients with high atherothrombotic risk. (D) Ischaemic endpoint in patients with low atherothrombotic risk. ATR, atherothrombotic risk; BARC, Bleeding Academic Research Consortium; CVM, cardiovascular mortality; HR, hazard ratio; MI, myocardial infarction.

study endpoints differed between TROPICAL-ACS strategies between both groups (Figure 5). Specifically, early de-escalation was not associated with an increase in ischaemic events, particularly not in the group of patients at high ATR. The finding that BARC 2–5 bleeding events did not differ between PFT-guided early P2Y₁₂ de-escalation and the control group in both low and high ATR might appear surprising. Nevertheless, one key parameter for distinguishing between low and high ATR in the present study was age. In an earlier pre-specified analysis of the TROPICAL-ACS trial, we identified a significant interaction between younger age and reduction of bleeding complications following a PFT-guided DAPT de-escalation.²⁷ Based upon the data of this specific *post hoc* analysis, a strategy of guided DAPT de-escalation might be safe and not associated with increased risk of ischaemic events also for patients with high ATR in whom de-escalation of antiplatelet treatment is deemed necessary. Recent network³⁶ as well as patient-level³⁷ meta-analyses investigating guided vs. standard selection of antiplatelet therapy showed a significant reduction in ischaemic and bleeding events. In the network meta-analysis performed by Galli *et al.*³⁶ including 11 RCTs and more than 20 000 patients, guided selection of antiplatelet therapy was associated with a reduction of major adverse cardiovascular events (MACEs) compared with standard therapy. In addition, guided antiplatelet therapy also resulted in a reduction of individual efficacy and safety outcomes, such as CVD, MI, stent thrombosis, stroke, and minor bleeding. The results were consistent,

independent of the test used for guided selection of therapy (PFT vs. genetic testing). Outcomes differed according to the strategy used. An escalation strategy was associated with a reduction in ischaemic events without an increase in bleeding and a de-escalation strategy with a reduction in bleeding, without an increase in ischaemic events.³⁶ In an accompanying editorial,⁹ the assets and drawbacks of guided DAPT strategies were summarized and discussed including its major advantage to potentially offer risk stratification even in the early post-procedural period, which is the most eventful period for patients after PCI. Similarly, Shoji *et al.*³⁸ showed in a network meta-analysis that a strategy of de-escalation of DAPT to clopidogrel or low-dose prasugrel was associated with significantly lower risk of bleeding while not increasing the risk of ischaemic events (including CVD, MI, stroke, and stent thrombosis) when compared with other DAPT strategies including clopidogrel, standard dose prasugrel, and ticagrelor.

Limitations

We are aware of some limitations of the present analysis. First, our findings were derived from a *post hoc* analysis of a randomized parallel-group open-label assessor-blinded non-inferiority trial. Therefore, the findings presented should be considered as exploratory rather than conclusive in nature, especially when interpreting single endpoints such as stroke or ST with small sample size in each subgroup and subsequently smaller



number of events. Nevertheless, the findings obtained in the present analysis may stimulate future prospective research with adequate statistical power to identify the best antithrombotic approach for patients at increased ATR. Second, several definitions for high ATR currently exist, which makes it difficult to compare our results with those of previous investigations. In the present analysis, we applied the definition used in the COMPASS trial since it was the most recent conducted trial in patients with high ATR. However, as the COMPASS trial only included patients with stable coronary artery disease, these criteria need further prospective evaluation in patients suffering from ACS. Due to the trial design and utilized potent antiplatelet therapy, the TROPICAL-ACS trial excluded patients with a history of recent stroke. Additionally, the case report form did not include left ventricular function as well as more detailed procedural characteristics to define PCI complexity¹³. Therefore, a more thorough classification of ATR based on PCI complexity cannot be applied. Additionally, only 370 (14.2%) patients in the TROPICAL-ACS trial were age > 70 years. Nevertheless, we were able to demonstrate high risk for ischaemic events in high-ATR patients according to the remaining criteria introduced by the COMPASS trial. Third, in the TROPICAL-ACS trial a PFT-guided de-escalation from potent P2Y₁₂ inhibitor prasugrel to clopidogrel was investigated. Therefore, our findings cannot be extrapolated to ACS patients treated with other antiplatelet drugs, e.g. ticagrelor. Neither can these results be transferred to DAPT de-escalation strategies that do not use platelet functional testing. The selection of timing of de-escalation was chosen based on previous evidence of the down-tapering of ischaemic events within the first weeks after ACS,³⁹ while others have used an unguided de-escalation after 30 days of ACS or a

very early genotype-guided de-escalation within 48 h of the initial presentation with ACS.⁴⁰ This variation makes the selection of a clear timing challenging. Additionally, there was a higher rate of use of GP GPIIb/IIIa antagonist, which is not reflective of current practice. Finally, for adjudication of MI, the 3rd universal definition of MI was used and not the more updated 4th universal definition of MI.

Conclusion

Using a risk score derived from the COMPASS trial in order to define increased ATR, we found that a PFT-guided early de-escalation from prasugrel to clopidogrel might be equally safe across varying levels of ATR and is not associated with an increase in ischaemic events. Further research is warranted to elucidate the safety and efficacy of de-escalation of antiplatelet therapy in patients at high ATR using various ATR scores.

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

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Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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